

Labwork 3: Logistic Regression

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Deep Learning – USTH

May 9, 2026

1 Implementation

```
def read_csv(path):
    x1s = []
    x2s = []
    ys = []
    f = open(path, "r")
    first = True
    for line in f:
        line = line.strip()
        if line == "":
            continue
        if first:
            first = False
            continue
        parts = line.split(",")
        x2 = float(parts[0])
        x1 = float(parts[1])
        y = float(parts[2])
        x1s.append(x1)
        x2s.append(x2)
        ys.append(y)
    f.close()
    return x1s, x2s, ys

def sigmoid(z):
    return 1 / (1 + math.exp(-z))

def predict(x1, x2, w0, w1, w2):
    z = w1 * x1 + w2 * x2 + w0
    return sigmoid(z)

def loss(w0, w1, w2):
    s = 0.0
    for i in range(N):
        y_hat = predict(x1s[i], x2s[i], w0, w1, w2)
        s = s - (ys[i] * math.log(y_hat) +
                  (1 - ys[i]) * math.log(1 - y_hat))
    return s / N
```

```

def train(w0, w1, w2, r, iters):
    for it in range(iters):
        for i in range(N):
            y_hat = predict(x1s[i], x2s[i], w0, w1, w2)
            e = y_hat - ys[i]
            w0 = w0 - r * e
            w1 = w1 - r * x1s[i] * e
            w2 = w2 - r * x2s[i] * e
    return w0, w1, w2

```

The training loop walks through all 19 data points each iteration and updates w_0 , w_1 , w_2 after every sample.

2 Results

The model is trained from initialization $w_0 = 0$, $w_1 = 1$, $w_2 = 2$ with learning rate $r = 0.05$ for 1000 iterations. Iteration results are shown below.

iter	w_0	w_1	w_2	J
1	-0.2021	0.0811	1.8676	0.8514
100	-3.9951	0.4903	2.0029	0.5007
200	-6.5077	0.7460	2.4038	0.3728
300	-8.3198	0.9323	2.7433	0.3178
400	-9.7412	1.0807	3.0276	0.2895
500	-10.9171	1.2053	3.2703	0.2730
600	-11.9229	1.3129	3.4823	0.2625
700	-12.8016	1.4072	3.6712	0.2553
800	-13.5804	1.4907	3.8420	0.2501
900	-14.2775	1.5651	3.9981	0.2460
1000	-14.9061	1.6318	4.1416	0.2428

The loss drops rapid during the first few hundred iterations and then go down slowly. After 1000 iterations the model converges to

$$w_0 = -14.9061, \quad w_1 = 1.6318, \quad w_2 = 4.1416, \quad J = 0.2428.$$

Evaluating with threshold $t = 0.5$ classifies 18 of 19 training data points correctly (94.7% accuracy); only 1 is misclassified.



Figure 1: Loan decision data and the decision boundary.